

Nikita Deshmukh

Cloud & DevOps Engineer

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PROFESSIONAL SUMMARY

Cloud & DevOps Engineer with 4+ years of experience designing, automating, and operating cloud-native platforms across AWS and Red Hat OpenShift. Skilled in Kubernetes, Terraform, Infrastructure as Code, CI/CD, Docker, observability, DevSecOps, AWS infrastructure, and PostgreSQL. Experienced in SRE practices and production incident response, with a strong focus on system reliability, GDPR-aligned security controls, cost optimization, and AI-driven operational automation.

EXPERIENCE

Software Engineer I | Volkswagen Group Digital Solutions India, Pune

Lastenheftassistant (LHA) - AI-Powered Requirements Engineering Tool | Aug 2022 - Present

- Designed and maintained Helm-based Kubernetes architecture for dev/test/staging/production with parameterized configs, dependency management, and zero-downtime releases.
- Built multi-stage Docker images for Python and Node.js with OpenShift-compliant security, non-root execution, and Docker Compose for local multi-container environments.
- Automated Aurora PostgreSQL schema migrations using Alembic and Kubernetes Jobs.
- Developed GitHub Actions CI/CD with SonarQube quality gates, Black Duck scanning, staging deployments, and production release approvals.
- Operated Prometheus, Grafana, Loki, Tempo, and OpenTelemetry for monitoring, logs, tracing, dashboards, and email alerts.
- Supported production incident triage and root-cause analysis using SRE practices.
- Managed TLS certificate lifecycle, secure routes, and Kubernetes secrets across OpenShift environments with GDPR-aligned data protection.
- Enforced outbound network security through custom proxy whitelist controls and domain-based egress restrictions.
- Reduced project costs by 35-40% through optimized pod configurations and pod creation strategy.
- Designed the LHA architecture migration path from OpenShift to an AWS-native solution (in progress).
- Built automated RCA workflows integrating LLMs to correlate infrastructure events, pod failures, deployments, and application logs.
- Collaborated in Agile Scrum using Jira across sprint planning, refinement, reviews, and retrospectives.

Core AWS Infrastructure & Platform Engineering

- Launched EC2 in custom VPCs with security groups, EBS, key pairs, private subnets, internet gateways, NAT gateways, and NACLs.
- Set up Application Load Balancers and autoscaling groups across multiple availability zones.
- Managed S3 lifecycle and bucket policies; created IAM users, groups, and roles with least-privilege access.
- Created reusable Terraform modules to provision AWS infrastructure (EC2, VPC, S3).
- Deployed containerized applications on Amazon ECS using Docker images in ECR.
- Configured CloudWatch logs, alarms, and dashboards for performance tracking and incident response.
- Designed a compute-optimized AVA chatbot on AWS Lambda and ALB, reducing infrastructure costs by 60-70%.

Hackathon Projects

- Built an AI-powered assistant for automotive PFMEA risk rating (i.mobilithon 5.0).
- Developed an automated system for comparing automotive part specifications using Python and OpenCV (i.mobilithon 4.0).

SKILLS

Cloud: AWS (EC2, S3, IAM, ECS), Azure (VNet, VMs, Key Vault), Red Hat OpenShift

Containers: Docker, Kubernetes, EKS, Helm

IaC & CI/CD: Terraform, YAML, GitHub Actions, Jenkins

Observability: Grafana, Loki, Prometheus, Tempo, CloudWatch, SNS

DevSecOps: SonarQube, Black Duck, Trivy, RBAC, IAM, OIDC, TLS/SSL, secrets management

Data: PostgreSQL (Aurora RDS), DynamoDB, Alembic

Automation: Python, Bash, Power Platform | AIOps: LLM, OpenAI, prompt engineering

EDUCATION

Bachelor of Engineering, Pune University - 9.52 CGPA

Aug 2018 - May 2022

AWARDS & TRAINING

- i.mobilitython 4.0 - Second Runner-up, Volkswagen Group Innovation Hackathon (Oct 2024 - Dec 2024).
- OpenShift with Kubernetes, VW Group internal training; on-site visit to Shanghai, China.

LANGUAGES

English and German (A2)